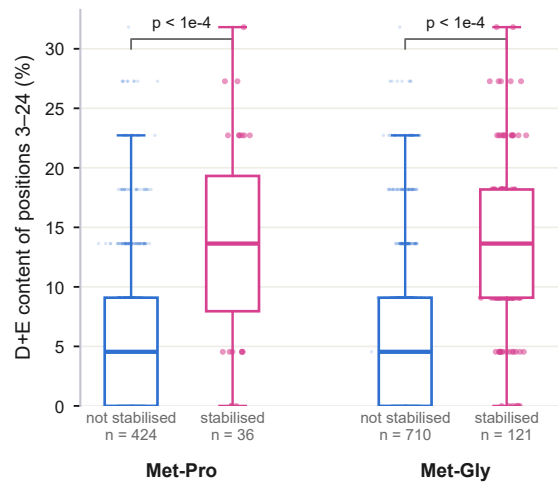


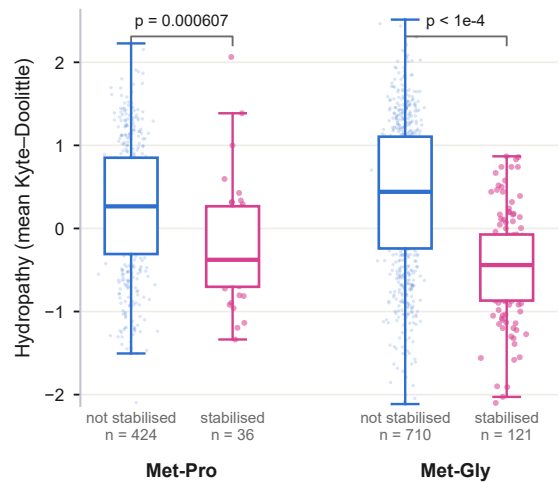
What separates stabilised peptides from unstabilised ones, with the N-terminal residue held fixed

A–C composition of positions 3–24, the part the motif does not fix · D–E the same comparison inside matched baseline-PSI bands, which rules out the stability ceiling as the explanation

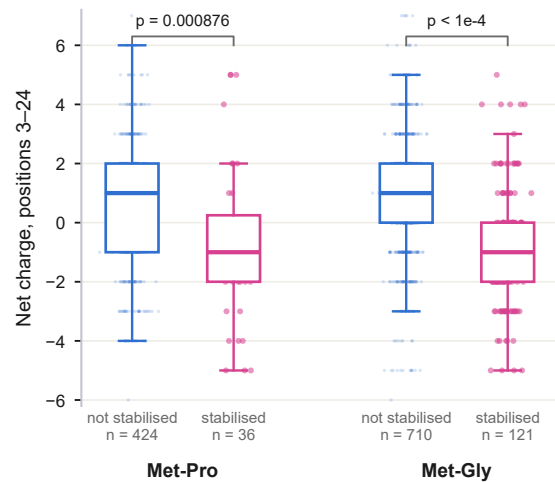
A



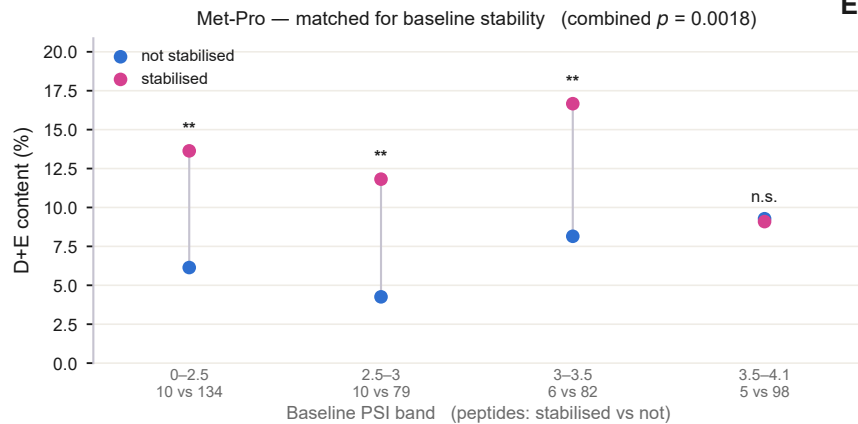
B



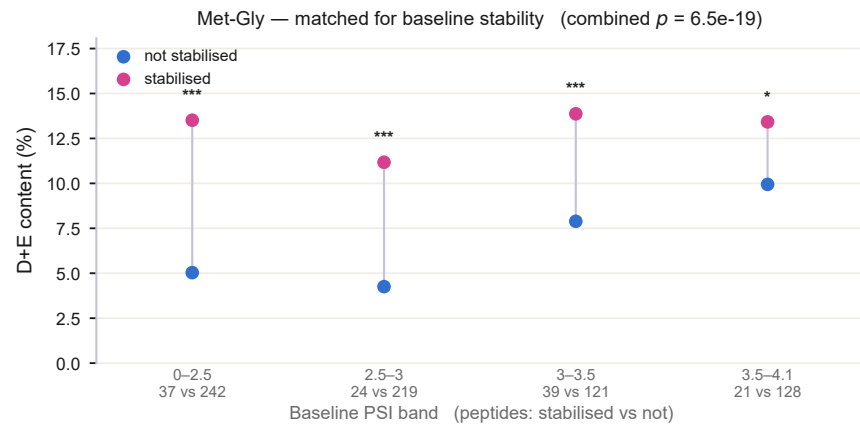
C



D



E



Stabilised = $\Delta\text{PSI} \geq 1.0$ with baseline $\text{PSI} \leq 4.0$ (Met-Pro: in any of ZYG11B KO, ZER1 KO, double KO; Met-Gly: double KO). Not stabilised = $\Delta\text{PSI} < 0.5$ everywhere measured. Boxes are median and quartiles, whiskers 1.5 IQR, every peptide drawn. Mann-Whitney throughout.

Among unstabilised Met-Pro peptides acidic content rises WITH baseline stability (Spearman $r = +0.22$, $p = 8e-06$), so the ceiling cannot manufacture panels A and D — it works against them. The same composition separates Met-Gly substrates more strongly than Met-Pro ones, so this is not proline-specific.